

Your baby's life support system

The placenta links your baby's blood supply to your own and carries out all the functions that your baby can't perform for himself. The placenta is rooted to the lining of your uterus and linked to the baby by the umbilical cord.

How the placenta develops

The placenta forms from cells in the embryo shortly after the egg implants in the lining of the uterus. Early placental growth is rapid, and at the beginning of the first trimester, the placenta is larger than the baby. However, the baby's growth catches up by 16 weeks, and by the end of pregnancy the baby is almost six times heavier. The final weight of the placenta is between 12 oz (350 g) and 1 lb 5 oz (700 g). Once its structure is complete at the end of the first trimester, it takes on many important functions for the rest of the pregnancy (see [Inside the placenta](#)).

Growth after the first trimester The placenta grows throughout the second trimester of pregnancy. By the third trimester, growth slows, but placental efficiency continually improves as extra villi (through which substances are exchanged; see [Inside the placenta](#)) grow and increase the available surface area of the placenta by nearly fourfold. Cell layers also become thinner so that substances can be exchanged efficiently.

Placental blood flow is massive and many of the changes in your circulation are designed to meet its needs. By term, a tenfold increase in placental blood supply results in a fifth of your circulation supplying the placenta with up to half a liter of blood (almost a pint) each minute.

The aging placenta The placenta ages toward the end of pregnancy, especially after 40 weeks. However, at least 60–80 percent of its function would need to be lost before there were any signs of problems with blood flow in the umbilical cord.

INSIDE THE PLACENTA

The placenta contains a huge network of tiny projections called chorionic villi that branch out from a thin membrane, the chorion, and contain fetal blood vessels. The chorionic villi are bathed in maternal blood within the intervillous space. Each villus is only one or two cells thick, which allows the transfer of gases and nutrients between the mother and the baby, while ensuring that the two circulations never come into direct contact. Through the process of diffusion, oxygen and nutrients such as glucose, your baby's primary source of energy, from the mother pass into the fetal circulation, and waste products from the baby are picked up and carried away in the mother's bloodstream. The chorionic membrane also acts as a protective barrier, preventing many harmful substances and infections from reaching the baby.